

IRELAND WEST AP KNOCK, Ireland

WMO: 039730

Lat: 53.910N

Lon: 8.818W

Elev: 203

StdP: 98.91

Time Zone: 1.00 (IRL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.0	-4.6	2.6	-0.1	-3.3	2.9	0.7	15.3	7.8	14.2	7.0	3.3	130	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	21.0	15.8	19.2	15.2	17.8	14.6	17.1	19.2	16.2	17.8	15.4	16.9	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.9	17.8	15.5	11.3	16.8	14.7	10.7	16.0	48.6	19.2	46.1	17.8	43.7	17.0	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.0	9.7	DB	-3.8	24.2	1.3	2.1	-4.7	25.7	-5.5	26.9	-6.2	28.1	-7.2	29.6	(4)
(5)			WB	-4.2	19.2	1.3	1.5	-5.1	20.3	-5.9	21.1	-6.7	21.9	-7.7	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	4.3	4.6	5.8	7.6	10.1	12.4	13.8	13.7	12.2	9.4	6.4	4.7	(6)	
	DBStd	4.21	2.64	2.43	2.50	2.41	2.62	2.34	2.07	1.65	1.91	2.22	2.46	2.83	(7)	
	HDD10.0	882	176	150	133	77	29	4	0	0	3	38	109	163	(8)	
	HDD18.3	3484	434	383	390	321	254	177	141	146	183	278	357	422	(9)	
	CDD10.0	443	0	0	1	6	34	77	119	113	70	18	2	1	(10)	
	CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0	(11)	
	CDH23.3	15	0	0	0	0	0	3	9	3	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	4.8	5.4	5.3	5.1	4.8	4.9	4.4	4.1	4.3	4.5	4.8	4.7	5.2	(14)	
(15) Precipitation	PrecAvg	1249	142	106	91	78	81	81	87	93	89	122	138	139	(15)	
	PrecMax	1541	224	254	159	158	159	197	177	184	182	204	283	319	(16)	
	PrecMin	926	29	25	36	25	16	31	40	9	12	40	61	52	(17)	
	PrecStd	142	49	58	32	29	38	38	33	42	43	50	47	64	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.2	14.5	17.1	21.7	23.1	24.9	21.8	19.7	15.9	13.5	12.0	(19)	
		MCWB	10.6	10.1	10.1	11.6	15.1	16.5	19.3	17.3	16.0	14.1	12.9	11.5	(20)	
	2%	DB	10.2	10.3	11.7	14.6	18.9	20.5	21.0	19.0	17.8	14.5	12.1	10.8	(21)	
		MCWB	9.7	9.6	9.2	10.4	13.7	15.0	16.4	15.6	15.0	13.4	11.6	10.4	(22)	
	5%	DB	9.5	9.5	10.4	12.9	16.5	18.3	19.0	17.6	16.5	13.7	11.1	10.0	(23)	
		MCWB	9.0	9.0	8.8	9.6	12.1	14.3	15.7	15.0	14.3	12.8	10.7	9.7	(24)	
	10%	DB	8.6	8.7	9.5	11.5	14.6	16.5	17.2	16.5	15.4	12.9	10.3	9.1	(25)	
		MCWB	8.1	8.0	8.2	9.1	11.2	13.3	14.6	14.3	13.7	11.9	9.8	8.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.7	11.1	12.5	16.0	17.2	19.3	18.3	17.1	14.9	13.1	11.7	(27)	
		MCDB	11.1	10.9	12.6	15.6	20.3	21.3	23.6	20.9	18.4	15.3	13.3	11.8	(28)	
	2%	WB	9.9	9.9	10.1	11.3	14.1	15.9	17.3	16.7	16.0	13.9	11.7	10.5	(29)	
		MCDB	10.1	10.2	10.9	13.3	17.6	19.0	19.7	17.9	16.9	14.2	11.9	10.7	(30)	
	5%	WB	9.2	9.1	9.4	10.4	12.9	14.8	16.2	15.8	15.0	13.1	10.8	9.8	(31)	
		MCDB	9.4	9.4	10.1	12.0	15.4	17.3	17.8	16.8	15.9	13.5	11.1	10.0	(32)	
	10%	WB	8.2	8.2	8.5	9.5	11.9	13.8	15.2	15.0	14.1	12.2	9.9	8.8	(33)	
		MCDB	8.5	8.6	9.2	11.1	13.9	15.9	16.8	16.0	15.0	12.7	10.2	9.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.8	5.7	6.5	6.8	6.5	6.1	5.7	5.7	5.1	4.5	4.2	(35)	
		MCDBR	5.1	5.1	6.5	8.2	9.6	9.0	8.5	6.9	6.7	5.3	4.9	5.3	(36)	
	5% WB	MCWBR	5.2	4.8	4.6	4.9	5.2	5.0	4.8	4.2	4.4	4.4	4.8	5.4	(37)	
		MCWBR	5.1	5.1	5.9	7.1	8.4	8.0	7.6	6.0	5.9	5.0	4.8	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.329	0.338	0.357	0.369	0.379	0.364	0.367	0.352	0.349	0.334	0.325	0.318	(40)	
		taud	2.239	2.322	2.313	2.306	2.316	2.389	2.388	2.441	2.445	2.433	2.284	2.202	(41)	
	Ebn at Noon		606	749	818	858	868	884	874	869	823	750	608	529	(42)	
		Edh at Noon	63	80	99	114	119	112	110	99	87	71	60	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.62	1.30	2.42	3.88	4.88	5.04	4.47	3.78	2.72	1.58	0.76	0.45	(44)	
	RadStd		0.06	0.12	0.21	0.33	0.40	0.43	0.30	0.30	0.18	0.13	0.07	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page